POAMS: LECTURE 2
HISTORICAL BACKGROUND II

1. Historical background and context II

In this second lecture, I will continue to present a potted history of POAMS, **'The Pope-Osborne Angular Momentum Synthesis'**, which provides a new approach to Newtonian mechanics, Relativity, physical phenomena at the quantum level and Physics in general. This is based on an observer-centred philosophy known as **Normal Realism.** As a reminder, this philosophy employs *Ockham's Razor*. This is the principle which in essence states that when presented with alternative explanations of physical phenomena, we should always opt for the simplest, the one with the fewest possible causes, assumptions or variables.

Again as a reminder, the collaboration between Viv Pope and I began in 1984, which led to our first joint paper, 'A new approach to Special Relativity' in 1987, which by interpreting *c* as simply as a *conversion factor* of observational distance to observational time, gives Viv's derivation of the time-dilation formula and the 'cone model'.

Having proposed a Normal Realist analogue of Special Relativity, in the early 1990s, we turned our attention to devising a Normal Realist account of General Relativity. Initial discussions centred around Newton's theory of gravity and Viv reiterated his idea that all natural motion has to be *orbital* with all such orbital motion being governed by the holistic law of *conservation of angular momentum*, rather than by some unseen theoretical *in-vacuo* 'gravitational force', as Isaac Newton (1642-1727) proposed.

For example, the reason why the Moon orbits Earth in the way that it does is not because 'the void' between them is in some kind of 'gravitational' tension. It is simply that the Moon's natural orbit around Earth is dictated by the conservation of its angular momentum relative to Earth and every other body. For the Moon to move in some way other than it does relative to Earth and vice versa – with the consequence that both bodies would have to change their motions relative to the fixed stars – some measure of real detectable force has to be applied to one or other of these bodies. And because that force requires the expenditure of energy, which is also conserved, such losses or gains in energy have to be accompanied respectively by corresponding gains or losses somewhere else.

I was initially sceptical concerning Viv's premise that conservation of angular momentum alone can determine natural orbits. I argued that some sort of equation of motion was also required. By jettisoning Newton's theoretical *in-vacuo* gravitational force, together with his inverse square law, the mathematics concerning orbital motion was made potentially much harder. However, I later came to realise that by introducing another degree of freedom into the derivation of the orbital equations, I could not only produce Newtonian orbits but also certain orbits predicted by General Relativity.

In order to concentrate on the crucial ideas, in the early 1990s we worked exclusively with natural and constrained *circular* orbits. The mathematics required to study such orbits is relatively straight-forward. A big step forward was to realise that since all types of angular momentum are holistically and instantaneously conserved, the presence of *spin* has an effect on natural orbital motion in a way that Newtonian theory does not predict. This realisation committed us to allowing for the Newtonian so-called 'gravitational constant' to be a variable. Extrapolating this idea down to the microphysical level, we were able to derive the correct parameters of the classical Bohr atom by introducing the concept of spin. We deduced the accepted values of the Bohr parameters by considering the spin kinetic energy of the 'orbiting particle' around the nucleus, without involving 'electrostatic charge', Coulomb force', or any other concept of that conventional kind.

As these ideas developed, we had discussions with Professor Tom E. Phipps Jr. and Professor H. Pierre Noyes. In due course, we were invited by the latter to submit another paper to *Physics Essays.* This we did and the paper appeared in 1995:

'Instantaneous Gravitational and Inertial Action-at-a-Distance', N. V. Pope and A. D. Osborne, *Physics Essays*, Vol 8, No 3 (1995), pp. 384-397.

In the second half of the 1990s Viv continued to publicise Normal Realism and POAMS mainly from a philosophical viewpoint by attending international conferences and having discussions with some participants at his home. As a consequence of this he was invited, with others, to edit a collection of international papers for a book for Nova Science:

'Instantaneous Action at a Distance: Pro and Contra', Andrew E. Chubykalo, Viv Pope and Roman Smirnov-Rueda, eds., Nova Science (2001), 459 pages.

During this period, my time was dedicated to finishing an undergraduate textbook on *Complex Variables* which I had been working on for several years and to developing the mathematics behind POAMS further by addressing general orbital motion, as opposed to just circular motion.

After the publication of the Nova Science book, Viv was approached by some of the contributors to the book, particularly Professor Peter Graneau, and asked to convene a meeting of those who were 'pro', particularly those who supported unmediated instantaneous action-at-a-distance. Consequently, this select group was invited to the meeting entitled 'Unmediated Instantaneous Action-at-a-Distance', subsequently known as the 'Swansea Workshop', held at the University of Wales, Swansea in July 2001. The success of the meeting was such that a second Workshop was arranged for April 2002. In the meantime, the debate was continued by email and letter. Viv gave talks on our work, which included the background to Normal Realism, the 'cone model' and instantaneous action-at-a-distance, the Pythagorean 'Quantum Accumulator', a derivation of the Balmer-Rydberg formula, the logical redundancy of both Newton's *in vacuo* 'gravitational force' and 'electric and magnetic forces', and the alleged 'expanding universe'. This was all under the general umbrella of moving from *atomism* to *holism* in modern Physics. I gave talks on the central ideas of POAMS and the full mathematical treatment of orbital motion under POAMS, including the effect of introducing spin at the macro- and micro-physical levels. This was the first time that the full mathematical treatment was presented.

As a result of these Swansea Workshops, it was decided to produce a book which contained a record of the often vigorous arguments, plus a record of some of the talks given. Viv and I, together with Professor Alan Winfield were the editors. All contributors were invited to contribute to the book but some, for various reasons declined. The book, published by Edwin Mellen

Press and containing a commendatory preface by Professor Clive Kilmister, appeared in 2005:

‘Immediate Distant Action and Correlation in Modern Physics: The Balanced Universe’, N. Vivian Pope, Anthony D. Osborne and Alan F.T. Winfield, eds., Edwin Mellen Press (2005), 306 pages.

During the time between the Swansea Workshops being held and this book coming to fruition, there were new developments in POAMS. These included what I consider to be one of my major contributions. In my Swansea Workshop talks, I showed that starting with conservation of momentum and assuming that the orbit in an isolated two-body system is closed, as observed in planetary motion, an equation of motion of the orbit can be deduced without the need for an *in-vacuo* ‘gravitational force’ and Newton’s inverse square law of gravitational attraction. Hence, it can be deduced that the planets orbit in ellipses as observed. However, I later realised that if the orbit is not assumed to be closed, once both conservation of momentum and time-dilation effects are taken into account, POAMS predicts, just in General Relativity, that the natural orbits of the planets are approximately ellipses with a small but detectable perihelion shift. These developments were included in Chapter Eleven of the book before publication.

In 2003, Viv invited me to attend the 25th annual international meeting of the Alternative Natural Philosophy Association (ANPA) in Cambridge, in order to present some of our ideas. He had been attending these meetings for a number of years. ANPA’s founding contributors included Professors Clive Kilmister and Pierre Noyes. I agreed to attend and gave a talk on POAMS, including the effect of spin on orbital motion at the September meeting. This was very well-received; the whole meeting being devoted to the study of Spin. This talk was duly published in the conference proceedings in May 2004:

‘The Pope-Osborne Angular Momentum Synthesis’, A. D. Osborne, *Proceedings of ANPA 25*, Cambridge, August 2003, pp. 22-43.

Previously, by attending various conferences, Viv had got to know Cynthia Whitney, editor-in-chief of *Galilean Electrodynamics*, a journal dedicated to non-traditional approaches to Physics. We had considered this journal an ideal outlet for our work and consequently, we had submitted a paper for publication dealing with POAMS and the effect of spin on orbital motion. This was duly published in 2003:

'An Angular Momentum Synthesis of "Gravitational" and "Electrostatic" Forces', A. D. Osborne and N. V. Pope, *Galilean Electrodynamics*, *GED-East*, Vol. 14 (2003), Special Issues No. 1, pp. 11-19.

I also attended the *ANPA* 27 meeting in Cambridge in 2005 in order to present my ideas of incorporating the perihelion shift effect into orbital motion by incorporating time-dilation. It was at this conference that I got to meet Cynthia Whitney.

During the first half of the 2000s, Viv and I were also working on books which were designed to encapsulate many of the ideas of Normal Realism and POAMS. Viv produced a mainly philosophical account of Normal Realism and POAMS in 2004 and published this work privately under 'Phi Philosophical Enterprises':

'The Eye of the Beholder, The Role of the Observer in Modern Physics', Viv Pope, Phi Philosophical Enterprises (2004), 103 pages.

In the meantime, I produced a book which encapsulated Viv and I's work to date from a very much mathematical viewpoint, with Viv contributing to the philosophical passages. The book includes chapters on orbital time dilation and the POAMS derivation of Schwarzschild space-time, together with a chapter on some aspects of astrophysics and cosmology. Again, this was published privately under Phi Philosophical Enterprises in 2007:

'Light-Speed, Gravitation and Quantum Instantaneity', Anthony D. Osborne and N. Vivian Pope, Phi Philosophical Enterprises (2007), 244 pages.

Viv continued to think deeply about questions of a philosophical nature and he wrote another book, together with his wife Mary, which takes the form of a philosophical dialogue. This was published privately by Phi Philosophical Enterprises in 2010:

‘Wondering in the Wilderness’, A. M and N. V. Pope, Phi Philosophical Enterprises (2010), 363 pages.

During the second half of the 2000s I wrote a further three papers of a mainly mathematical nature for *Galilean Electrodynamics,* with contributions from Viv. The first of these covered POAMS and orbital time dilation, leading to my derivation of Schwarzschild space-time. This was some of the material which had been covered by our privately published book, ‘Light Speed…’:

‘Orbital Time Dilation’, Anthony D. Osborne and N. Vivian Pope, *Galilean Electrodynamics*, Vol. 19, No. 2 (2008), pp. 23-29.

The second paper was also based on material from ‘Light Speed…’ and covered our new approach to Special Relativity based on the philosophy of Normal Realism:

‘A Neo-Phenomenalist Alternative to Special Relativity Theory’, Anthony D. Osborne and N. Vivian Pope, *Galilean Electrodynamics*, Vol. 19, No. 4 (2008), pp.63-67.

We had to wait until 2012 for the publication of the third and final of this trio of papers. This concerned Viv and I’s work on POAMS and the cosmological redshift, which was based upon the relevant material in ‘Light Speed…’:

‘POAMS and the Cosmological Redshift’, N. Vivian Pope and Anthony D. Osborne, *Galilean Electrodynamics*, Vol. 23, No. 6 (2012), pp. 103-108.

By the time this paper was published, sadly Viv had died and this volume contains a tribute to Viv Pope by Cynthia Whitney.

Viv continued to develop the philosophy behind POAMS almost to the time of his death. In around 2010, he had the idea of writing a comprehensive account of our work from a philosophical standpoint in the form of a book. Although he was the major contributor to this book, I contributed to some parts and was heavily involved in the editing. Viv finally settled on the title ‘Einstein’s Lost Legacy’ for the book. Viv also had the idea of getting a DVD produced on our work, with me contributing some short lectures on the mathematics. The DVD was filmed over a number of weeks and appeared in 2011:

'POAMS: A Life's Work, The Pope-Osborne Angular Momentum Synthesis', N. V. Pope (and A. D. Osborne), DVD, Velica Films (2011).

Although Viv got to see the finished DVD, he never lived to see the planned book published. I continued with the editing and the production of the final version of the book after his death in 2012, with the help of Penny Gower and Viv's daughter Alison.

Over the years, Viv and I received a lot of correspondence concerned with Normal Realism and POAMS. Not surprisingly, most of the correspondence from physicists who were firm believers in the currently accepted theories was very negative about our work. From time to time, we did receive very favourable comments and support for our work, particularly from teachers of Special Relativity and related material. Then in July 2013, I received an email from Chris Milam, who at the time described himself as a financier and property developer. Chris was hugely supportive of our Normal Realist approach and POAMS, and this signified the beginning of an on-going exchange of ideas by email and meetings.

Chris arranged to visit the UK, on his way back to the USA from France, in April 2014 in order to discuss Viv and I's work with me. We met at Keele University and in the local area. I invited Alison to a couple of these meetings, so that she could talk about her dad to Chris. Chris expressed a particular interest in POAMS at this time and the possible effect of spin on orbital motion and weight. He gave us the very welcome news that he was keen to generously fund experiments to support POAMS and its predictions concerning spin.

In early 2015 Chris's company, Quantum Machines, contracted with the Marshall Space Flight Center: Advanced Propulsion Laboratory (MSF CAPL) in Huntsville, Alabama, which was established by NASA. Quantum Machines channelled substantial funding into MSFCAPL to pursue the design, development and operation of experiments based on POAMS, with MSFCAPL providing its physical plant, facilities and experience.

Meanwhile, Chris investigated the publication of ‘Einstein’s Lost Legacy’ and decided the best way forward was to publish directly with Amazon, to make it available as a Kindle eBook and by a print on demand basis. Chris generously funded the publication of the book, using an intermediate company, Watershed5, for the submission process. This turned out to be a very lengthy process due to necessary reformatting and other matters. The book finally appeared in April 2016, with Chris and Preston Carter, Chief Technology Officer of Quantum Machines, and Penny Gower and others providing very positive reviews:

‘Einstein’s Lost Legacy’, N. Vivian Pope and Anthony D. Osborne, Amazon (2016), 259 pages.

After initial publication, further corrections to typographical and publishing errors were made before the final version appeared in June 2017.

In 2015, I was invited by Chris to travel to Huntsville in order to present the main ideas of POAMS to his colleagues at MSFC. I made the trip in September of that year and presented several short lectures, which were filmed at conference facilities in the Westin Hotel. Chris, Preston Carter, together with Richard Eskridge, Michael Nelson and Michael Schoenfeld, all experimental scientists at MSFC, plus a couple of research students were present. The lectures were well-received but I believe that Richard and the two Michaels were initially sceptical about POAMS predictions concerning spin. Nevertheless, a large number of different experiments and were devised and carried out at MSFC by Richard and the two Michaels beginning in early 2016.

I was invited back to Huntsville in July 2016 by Chris and his team to discuss the experiments, which involved rotating gyroscopes. I was able to visit the MSFC during my visit to see the experimental apparatus being used at that time. We were all very disappointed at the time to find that the data from the evolved rotor testing program did not appear to show any notable effects due to spin. This was a major set-back. We discussed the use of materials for which the nuclear spin is highly correlatable as a possible way forward. Then in October I received the very welcome news that the next experimental rig had clearly demonstrated weight change due to spin. The predictions of POAMS were vindicated at last. Richard and both Michaels were now firm believers in the predictions of POAMS.

Since that time, we have very sensitive weighing scales and gyroscopes which can be imparted with a significant spin using an electric motor, and these provide a demonstration of the effect of spin on weight. The comprehensive experimental work carried out by Richard and the two Michaels at MSFC was written up as a NASA technical report:

'A Study of the Pope-Osborne Angular Momentum Synthesis Theory (POAMS) Including a Mathematical Reformulation and Validation Experiment', R. H. Eskridge, M. A. Nelson, and M P. Schoenfeld, NASA/TM-20205010911, Marshall Flight Center, Huntsville, Alabama.

Having presented the most important steps in the history of Normal Realism and POAMS, in the next lecture I will move on to talk about the philosophical basis of Viv and I's work.

Anthony D. Osborne, March & April 2023.